Neel Gajare

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Education

University of California, Berkeley — Berkeley, CA
B.S. EECS (Electrical Engineering & Computer Science) — *Dec 2026*

Relevant Coursework (3.9 GPA): Scalable AI, Advanced Topics in Computer Systems, Advanced Algorithms, Computer Architecture, Machine Learning, Operating Systems, Computer Security, Database Management, Quantum Computing, Optimization Models in Eng, Circuit Design, Discrete Math & Probability, Electricity & Magnetism, Mechanics, Competitive Programming, Digital Design and Integrated Circuits, FPGA Lab, Advanced LLM Agents

Societies: Space Technology @ Cal, Cloud Computing @ Cal, Codify, Eta Kappa Nu (EECS Honor Society), Sky Computing Lab

Experience

ML Intern @ Apple — May 2026 – Aug 2026
Apple — *Santa Clara, CA*

- Worked on developing agentic AI systems and training neural network models such as GNNs to accelerate and optimize physical design workflows such as placement & routing and timing path diagnosis for next-generation Apple silicon.

AI Researcher — Feb 2024 – Present
Berkeley AI Research Lab (BAIR) & Sky Computing Lab — *Berkeley, CA*

- Sky Computing Lab: Working under Professor Alvin Cheung on LLM-driven kernel optimizations on tensor accelerators which utilizes a beam search over LLM-generated optimization plans, pruned by correctness tests and cycle-accurate performance feedback. Achieved 1.4× to 3× performance over various expert hand-tuned schedules on TPU v6. Previously worked on using thought trees to predict reasoning models accuracy on coding tasks. Paper published in COLM.
- BAIR: Worked under Professor Kurt Keutzer on Generative AI project pertaining to low resource machine translation and linguistic segmentation. Utilized Langchain, Chroma, Milvus to implement efficient RAG techniques on billion-token corpus to improve translational model. Integrated vector database to backend system and API with FastAPI. Implemented agentic workflows for paragraph segmentation techniques to enhance retrieval system.

SWE Intern @ Amazon — May 2025 – Aug 2025
Amazon — *Sunnyvale, CA*

- Designed low-latency traffic analysis workflows to detect and identify unique non-volumetric and agentic bot signatures affecting Amazon's retail websites and backend services.
- Utilized AWS tools such as Athena, Sagemaker AI, Cloudwatch, EC2, and internal tools to automate manual traffic spike investigation time for oncall engineers by >90% and reduce compute costs by >85% on average.

Stanford Research Intern — May 2022 – Feb 2024
Stanford University — *Palo Alto, CA*

- Worked in Stanford's Crustal Deformation Lab under Professor Paul Segall. Researched complex basaltic magma chamber geometries using AI.
- 1st Author of abstract published in American Geophysical Union (AGU) conference proceedings, presented in AGU Fall Meeting. Utilized Gmsh for 3-D mesh generation to model magma chambers, Matlab for simulation of surface deformation and Bayesian optimization techniques.

Awards

- Apple FPGA Digital Design Contest - 1st Place — May 2025
- Nvidia National Merit Scholarship — April 2023
- St. Francis de Assisi Volunteering Award — May 2022

Projects

Pintos Operating System | *C, x86, GDB* — Jan 2026 - May 2026

- Extended the Pintos kernel with system calls, user-memory validation, and process lifecycle management, including parent-child synchronization and per-process file descriptor tables.
- Implemented kernel-backed user threads with create/join/exit operations, per-thread stacks, and locks and semaphores to coordinate shared resources and process teardown.
- Built an extensible file system with indexed inodes, hierarchical directories, and a 64-entry write-back buffer cache using clock eviction and per-inode locking for concurrent file access.

RISC-V CPU on FPGA | *Verilog* — Jan 2025 – May 2025

- Implemented 5-stage pipelined RISC-V CPU on FPGA, with UART tethering, BIOS functionality, synchronous memories.
- Optimized CPU to run on 125 MHz clock frequency and 1.06 average CPI on various benchmarks by resolving pipelining hazards, designing a pipelined hybrid Gshare branch predictor + branch target buffer + return adress stack.
- Achieved highest figure-of-merit score in Berkeley's digital design competition sponsored by Apple.

Technical Skills

Languages: Python, C, C++, Verilog, SQL, RISC-V Assembly, JavaScript
Frameworks/Tools: vLLM, Triton, LangGraph, AWS EC2, Athena, Sagemaker AI, Docker, Kubernetes, Trainium, LangChain, Pytorch, Vivado, Keras, OpenCV, Gmsh, Numpy, Jax, MongoDB, OpenAI, HuggingFace, Chroma, FastAPI, Firebase, Valgrind, Streamlit, Milvus, GDB, TensorRT, PrimeTime, NSight Systems/Compute, SGLang, CUDA, Pallas
Skills: AI/ML, Chip Design, Computer Architecture, Algorithms, Data Processing/Analysis, EDA, NLP